LABORATORY RECORD

Course: Full Stack Web Development
Course Code: 23CM4121
Experiment No.: 1(2)
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1. TITLE

HTML5 Graphics and CSS3 Web Styling

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2. AIM

To design and develop a styled webpage using **HTML5 and CSS3**, incorporating HTML5 Canvas graphics, navigation, content cards and various CSS3 styling properties.

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3. OBJECTIVE

The objectives of this experiment are:

- To understand the use of HTML5 and CSS3 in webpage development.
- To create a structured webpage with a header, navigation bar, content cards and footer.
- To apply CSS3 properties for styling different webpage components.
- To create a gradient background using CSS3.
- To apply borders, border-radius, margins, padding and box shadows.
- To use Flexbox for arranging multiple content cards.
- To implement hover effects and CSS transitions.
- To understand the HTML5 <canvas> element.
- To draw a rectangle, circle and line using Canvas.
- To display text on the Canvas using JavaScript.

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4. THEORY

HTML5 provides several features for creating modern web pages, including the **Canvas API** for drawing graphics directly in the browser. The <canvas> element provides a drawing area that can be controlled using JavaScript.
CSS3 is used to style and enhance the appearance of HTML elements. It provides properties for backgrounds, fonts, borders, spacing, shadows, transitions, transformations and layouts.
In this experiment, a webpage titled **"My Styled Webpage"** is created using HTML5 and CSS3. The webpage contains a header, navigation bar, three content cards, a Canvas section and a footer.
CSS3 properties such as linear-gradient(), background-color, border-radius, box-shadow, padding and margin are used to improve the visual appearance. **Flexbox** is used to arrange the three cards horizontally. Hover effects are applied to navigation links and buttons using the :hover pseudo-class, transition and transform.
The HTML5 Canvas API is used to draw a rectangle, circle and line. Text is also displayed inside the Canvas using JavaScript methods.

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5. ALGORITHM / PROCEDURE

- Create an HTML5 document using the <!DOCTYPE html> declaration.
- Create the <head> section and specify the character encoding and viewport.
- Add the webpage title.
- Define CSS styles inside the <style> element.
- Set a gradient background and font for the webpage.
- Create a header containing the webpage title and description.
- Create a navigation bar with Home, Cards and Canvas links.
- Create a container for three content cards.
- Add headings, paragraphs and buttons to each card.
- Use CSS Flexbox to arrange the three cards.
- Apply borders, rounded corners, padding and box shadows to the cards.
- Add hover effects to navigation links and buttons.
- Create a Canvas element with specified width and height.
- Obtain the Canvas element using document.getElementById().
- Obtain the 2D drawing context using getContext("2d").
- Draw a rectangle using fillRect().
- Draw a circle using arc() and fill().
- Draw a line using moveTo(), lineTo() and stroke().
- Set the font and display text using fillText().
- Create a footer containing copyright information.
- Save the HTML file and open it in a web browser.
- Verify the CSS styling, cards, hover effects and Canvas graphics.

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6. PROGRAM

```
<!DOCTYPE html>
<html lang="en">

<head>
   <meta charset="UTF-8">
   <meta name="viewport" content="width=device-width, initial-scale=1.0">
   <title>My Styled Webpage</title>
   <style>
      body{
         background-image: linear-gradient(to right,rgb(167, 192, 205) 30%, rgb(169, 189, 142)
70%);
         font-family: Arial, Helvetica, sans-serif;

      }
      header{
         background-color: cadetblue;
         padding: 10px;
         border-radius: 10px;
         box-shadow:5px 5px 10px gray;
         text-align: center;
      }
      nav{
         background-color: rgb(126, 178, 195);
         padding: 20px;
         margin: 5px;
         border-radius: 10px;
         text-align: center;

      }
      nav a{
         background-color: aquamarine;
         color: black;
         padding: 7px;
         margin: 20px ;
         text-decoration: solid;
         border-radius: 15px;
      }
      nav a:hover{
         background-color: rgb(164, 255, 225);
      }
      .threecards{
         display:flex;
         justify-content:space-around;
      }
```

```
        .card h1{
            color: rgb(85, 2, 93);
        }
        .card{
            width:28%;
            background: azure;
            border-radius:10px;
            padding:20px;
            margin:10px;
            box-shadow:5px 5px 10px gray;
            text-align:center;
        }
        .card button{
            background-color: aquamarine;
            color: black;
            padding: 7px;
            text-decoration: solid;
            border-radius: 15px;
            cursor: pointer;
            transition: all 0.3s ease;
        }
        .card button:hover{
            background-color: rgb(164, 255, 225);
            transform: scale(1.05);
        }
        canvas{
            display: block;
            margin:20px auto;
            border: 2px solid black;
            border-radius: 10px;
        }
        footer{
            text-align: center;
        }
    </style>
</head>

<body>
    <!--Part B-->
    <header>
        <h1>Styled Web Page</h1>
        <p>A style page using html and css</p>
    </header>
    <hr>
```

```html
<nav>
   <a href="#home">Home</a>
   <a href="#cards">Cards</a>
   <a href="#canvas">Canvas</a>
</nav>
<hr>
<!--Part B cont-->
<div class="threecards" id="cards">
   <div class="card">
      <h1>Card1</h1>
      <p>Hello Welcome</p>
      <button type="submit">Submit</button>
   </div>
   <div class="card">
      <h1>Card2</h1>
      <p>You are beautiful</p>
      <button type="submit">Submit</button>
   </div>
   <div class="card">
      <h1>Card3</h1>
      <p>Welcome to Jays Web</p>
      <button type="submit">Submit</button>
   </div>

</div>
<hr>
<!--Part A-->
<h1 align="center" style="color: indigo;">Canvas</h1>
<canvas id="newcanvas" width="500" height="300"></canvas>

<script>
   const canvas = document.getElementById("newcanvas");
   const can = canvas.getContext("2d");

   // Rectangle
   can.fillStyle = "green";
   can.fillRect(20, 20, 120, 80);

   // Circle
   can.beginPath();
   can.arc(250, 60, 40, 0, 2 * Math.PI);
   can.fillStyle = "orange";
   can.fill();
```

```
        // Line
        can.beginPath();
        can.moveTo(20, 150);
        can.lineTo(250, 150);
        can.strokeStyle = "red";
        can.lineWidth = 3;
        can.stroke();

        // Text
        can.font = "24px Arial";
        can.fillStyle = "black";
        can.fillText("MY Web", 20, 230);
    </script>
    <hr>
    <footer>
        <p>&copy; 2026 Anits College Info Portal. All Rights Reserved.</p>
    </footer>
</body>

</html>
```

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7. CODE EXPLANATION

HTML and CSS Code

<header>

Defines the header section of the webpage.

<nav>

Creates the navigation bar containing links to different sections.

.threecards

Defines the container for the three content cards.

display: flex;

Uses CSS Flexbox to arrange the cards in a flexible layout.

justify-content: space-around;

Distributes the cards with space around them.

.card

Defines the appearance and layout of each content card.

background: azure;

Sets the background color of the cards.

border-radius: 10px;

Creates rounded corners for the cards.

box-shadow
Adds a shadow effect around the header and cards.
linear-gradient()
Creates a gradient background by smoothly transitioning between two colors.
:hover
Applies special styling when the mouse pointer moves over an element.
transition
Creates a smooth change between the normal and hover states.
transform: scale(1.05)
Slightly enlarges a button when the user hovers over it.
canvas
Defines the styling applied to the HTML5 Canvas element, including its border, margin and rounded corners.
Canvas and JavaScript Code
document.getElementById("newcanvas")
Selects the Canvas element from the webpage using its ID.
getContext("2d")
Obtains the 2D drawing context required for creating graphics.
fillStyle
Specifies the color used to fill a shape or text.
fillRect(20, 20, 120, 80)
Draws a filled rectangle at the specified position with the given width and height.
beginPath()
Starts a new drawing path.
arc()
Creates a circular or arc-shaped path.
fill()
Fills the current path with the selected fill color.
moveTo()
Specifies the starting point of a line.
lineTo()
Specifies the ending point of a line.
stroke()
Draws the outline of the current path.
lineWidth
Specifies the thickness of the line.
font
Sets the font size and font family used for Canvas text.
fillText()
Displays text at a specified position on the Canvas.

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8. TEST CASES AND EXECUTION DATA

I would not just write three random test cases. I would actually execute them and record the results.

Test Case	Input / Action	Expected Result	Actual Result	Status
TC-01	Open the HTML file in a browser	Styled webpage should be displayed	Webpage displayed successfully	Pass
TC-02	Verify the header	Header with title and description should be displayed	Header displayed correctly	Pass
TC-03	Verify navigation bar	Home, Cards and Canvas links should be displayed	Navigation displayed correctly	Pass
TC-04	View the Cards section	Three cards should be displayed	Three cards displayed	Pass
TC-05	Hover over navigation links	Hover styling should be applied	Hover effect displayed	Pass
TC-06	Hover over card buttons	Button should change appearance and scale	Hover effect displayed	Pass
TC-07	View webpage background	Gradient background should be visible	Gradient displayed correctly	Pass
TC-08	View Canvas section	Canvas area should be displayed	Canvas displayed correctly	Pass
TC-09	Execute Canvas drawing code	Rectangle, circle and line should be drawn	Shapes displayed successfully	Pass

Test Case	Input / Action	Expected Result	Actual Result	Status
TC-10	Execute Canvas text code	Text MY Web should be displayed	Text displayed successfully	Pass
TC-11	Verify footer	Footer should be displayed at the bottom	Footer displayed correctly	Pass

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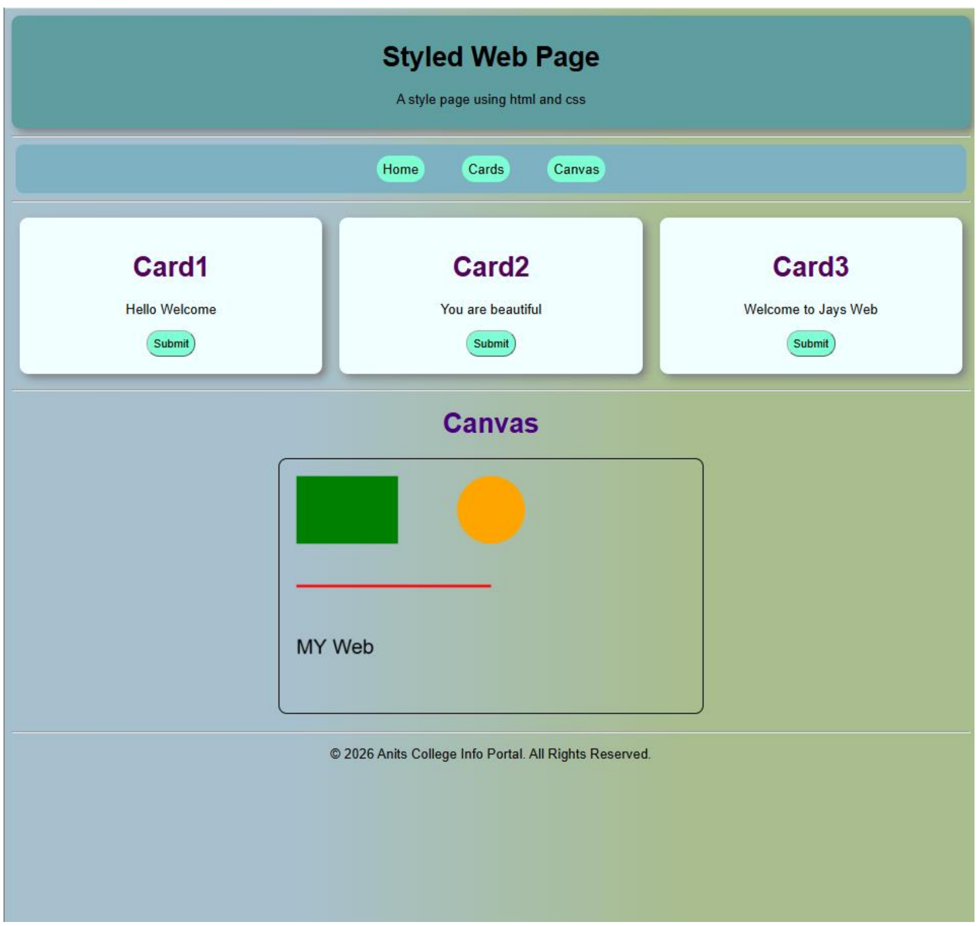
9. OUTPUT



10. RESULT

Thus, the **HTML5 Graphics and CSS3 Web Styling** webpage was successfully designed and implemented. The webpage contains a header, navigation bar, three styled content cards, hover effects and a footer. HTML5 Canvas was successfully used to draw a **rectangle, circle, line and text**, while CSS3 properties were used to enhance the overall appearance and layout of the webpage.